Critical Appraisal of Scientific Papers

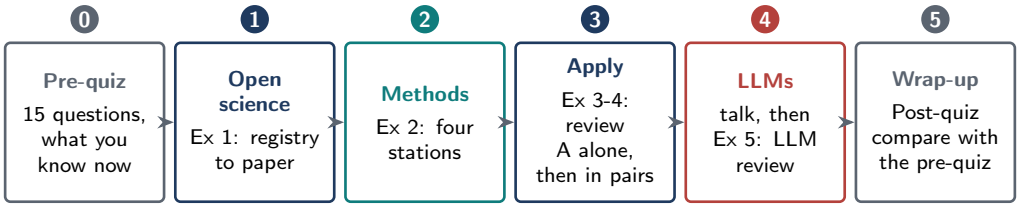
Reading a clinical paper the way a good peer reviewer does, with and without AI

Imad El Badisy, PhD

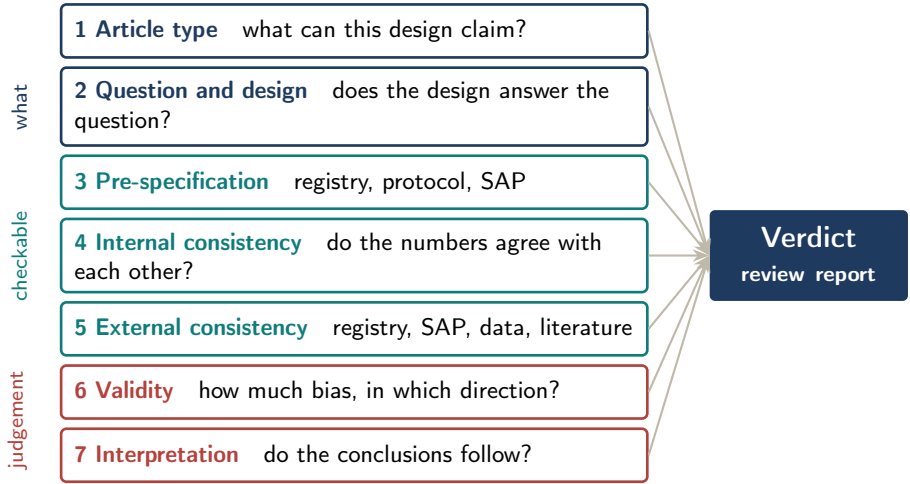
CM6RI

One-day workshop

The day: learn on two real papers, apply on one



One tool all day: seven layers, then a verdict



Internal consistency: the numbers in the paper agree with each other. **External consistency**: the paper agrees with its registry, protocol, SAP and other reports. **SAP**: statistical analysis plan.

Checklist: handouts/checklist-7-layers.md

Pre-quiz

individual

10

minutes

START NOW

Instructions and answer grid: pack p. 2-3

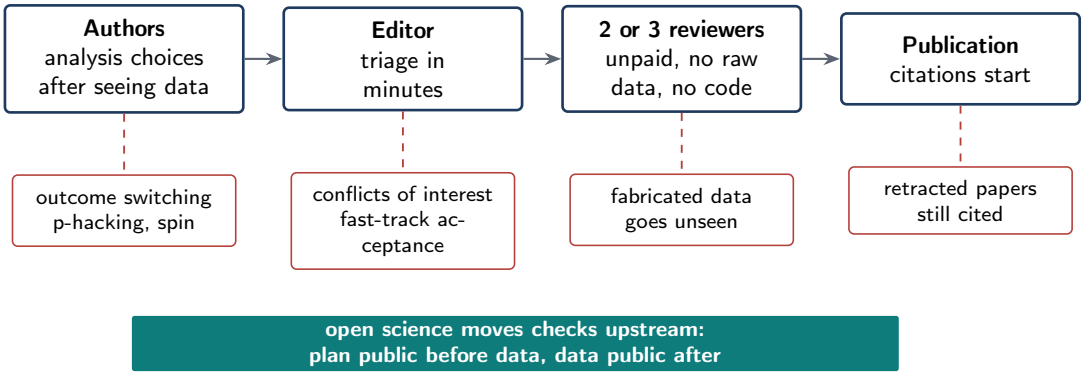
15 questions, one answer each.

STOP when time is up → next: block 1, open science

1 Open science and pre-specification

What was promised, what was published, and who can check. Layers 1, 2, 3 and 5.

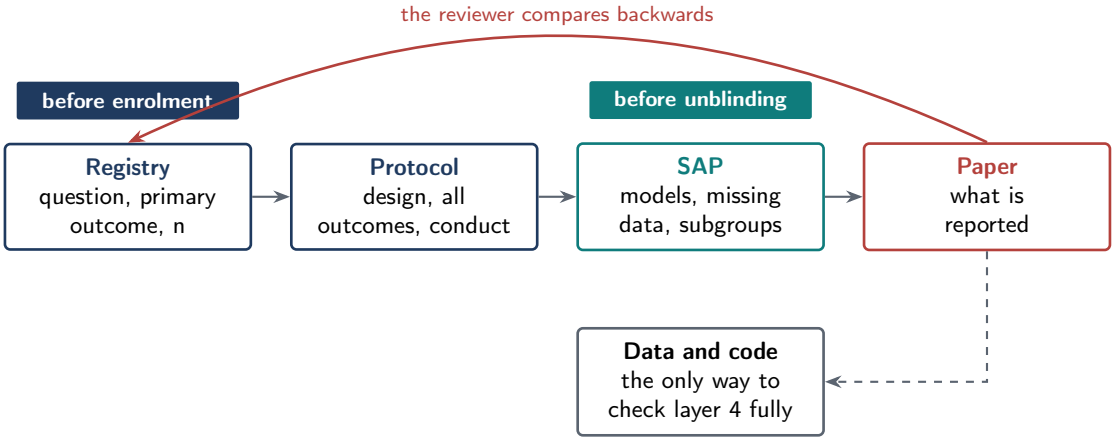
Where publishing breaks



Outcome switching: reporting a different primary outcome from the one registered. **p-hacking**: trying analyses until one gives $p < 0.05$. **Spin**: wording that makes results look better than they are. **Fast-track**: acceptance after a shortened review (Gautret: accepted the day after submission).

Cases used today: Gautret 2020 (fast-track, retracted 2024); Elgazzar 2020 (withdrawn, fabricated records).

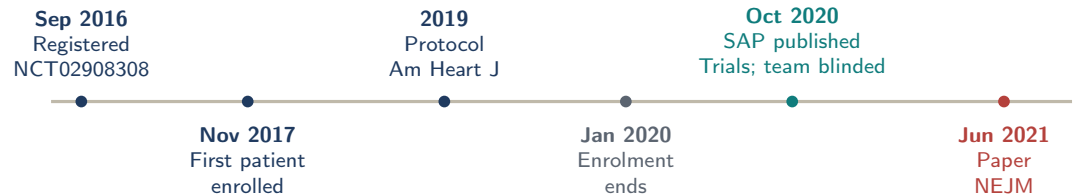
Four documents, each locks something before the data are seen



Registry: public record of a study (e.g. ClinicalTrials.gov). **SAP**: statistical analysis plan, the analyses written in detail before the data are seen. **Unblinding**: the moment the team learns who received which treatment.

SPIRIT 2025 (protocols), CONSORT 2025 (reports), ICH E9(R1) (estimands and SAP).

TTM2: a paper trail you can follow



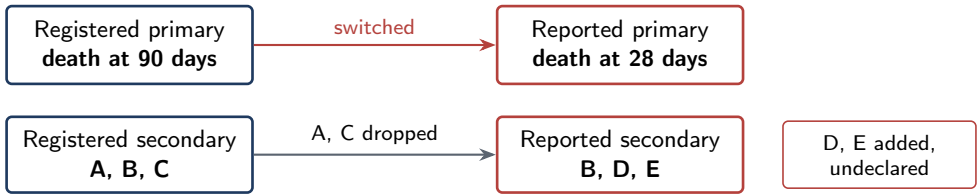
Primary outcome
death at 6 months:
same in all four documents

Sample size 1200 to 1900
for TAME co-enrolment,
before any results

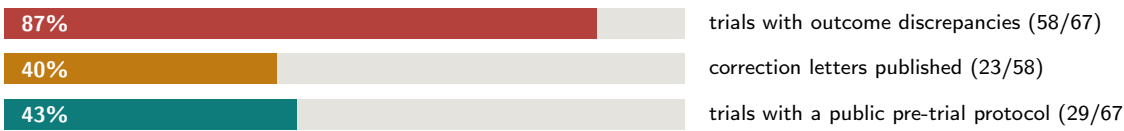
One detail
SAP relative reduction 13.6%,
paper says "15%"

TAME co-enrolment: patients could also enter the TAME trial; the sample size was raised so both trials kept their power. **Relative reduction**: (control risk − treated risk) / control risk.

Outcome switching: the registered outcome quietly becomes another



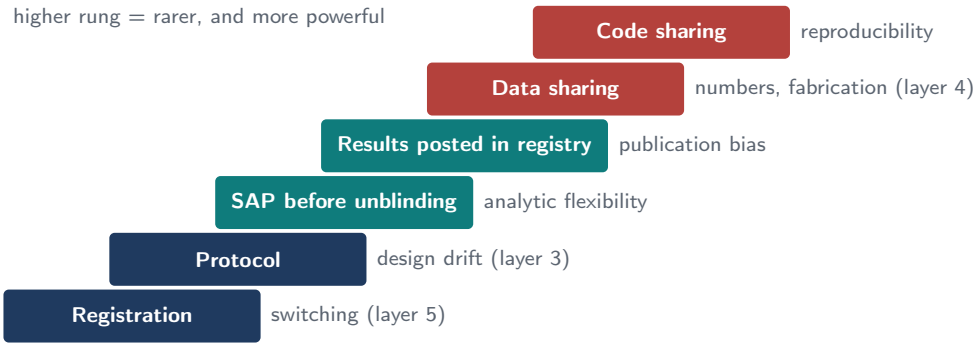
COMPare: all 67 trials over 6 weeks in NEJM, Lancet, JAMA, BMJ, Annals



Discrepancy (COMPare): a registered outcome not reported, or a new outcome added without saying so. **Correction letter**: a letter to the journal pointing out the discrepancy.

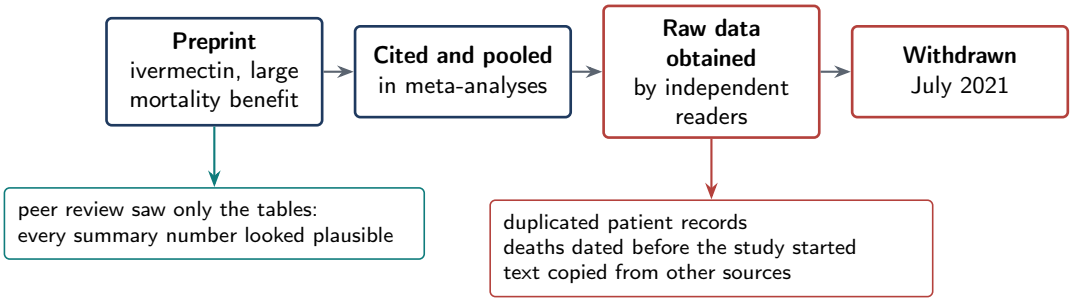
Goldacre et al. COMPare. Trials 2019;20:118 (materials/references/method-goldacre-compare-2019.pdf).

Open science is a ladder: each rung makes one more check possible



Analytic flexibility: many defensible analyses exist; choosing after seeing the data favours the one that "works". **Publication bias**: negative results are published less often. **Reproducibility**: same data and code give the same numbers.

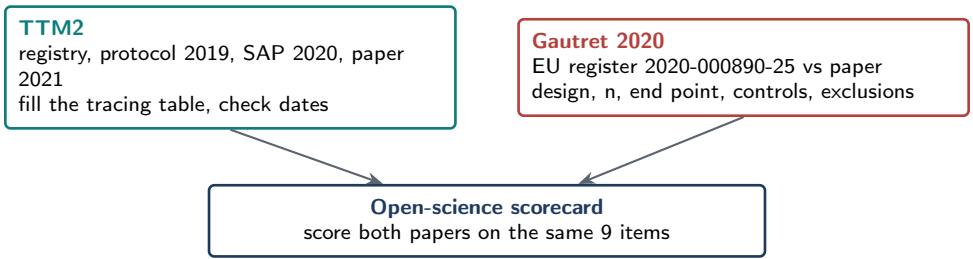
Elgazzar 2020: only the raw data could show it



Preprint: a manuscript posted online before peer review. **Meta-analysis**: a pooled estimate from several studies; one fabricated study can shift it. **Raw data**: one row per patient, before any summary.

Research Square withdrawal note, 14 July 2021 (materials/references/article-elgazzar-2020-withdrawn.pdf).

Exercise 1 Trace two papers from promise to publication



Pairs 50 minutes + 10 debrief pack p. 4-5

Part A TTM2 (25 min), Part B Gautret (20 min), Part C scorecard (5 min).

Exercise 1 — Registry to paper

pairs, laptops open

50

minutes

START NOW

Instructions and answer grid: pack p. 4-5

Part A (25 min): fill the TTM2 table, answer Q1-3.
Part B (20 min): fill the Gautret table, answer Q4-6.
Part C (5 min): score both papers, answer Q7.
TTM2 protocol and SAP: materials/workshop/

STOP when time is up → next: debrief (10 min); break

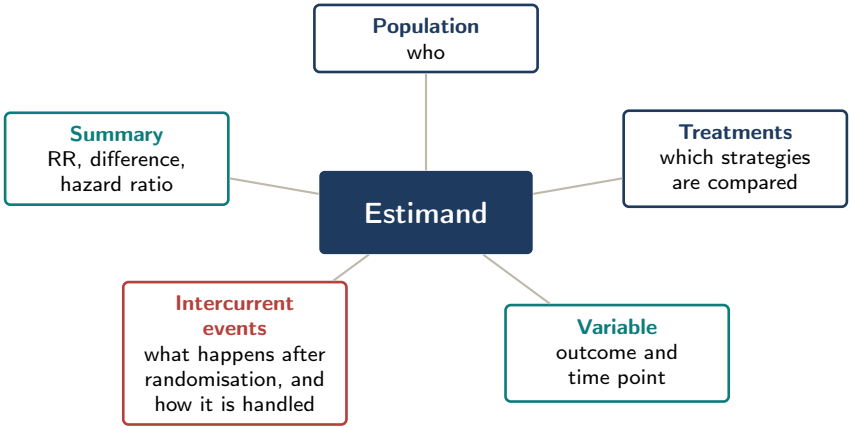
Break — 15 minutes

Next: block 2, statistical methodology on real papers

2 Statistical methodology on real papers

Estimand, confidence intervals, forensics, bias, spin. Layers 2, 4, 6 and 7, on TTM2 and Gautret.

The estimand: five attributes that define the question



Estimand: the exact quantity the study sets out to estimate. **Intercurrent event**: something after randomisation that changes how the outcome can be read (death, stopping treatment, rescue therapy). **Summary measure**: the number that compares the groups.

ICH E9(R1) addendum on estimands, 2019.

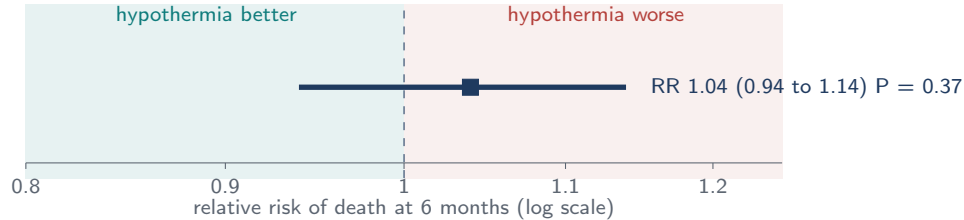
Same words, different questions: TTM2 vs Gautret

	TTM2	Gautret
Population	comatose adults after OHCA	inpatients, one centre; controls from other centres
Treatments	33 °C vs normothermia	HCQ (± AZ) vs no HCQ, not assigned
Variable	death by 6 months	PCR negative at day 6 (surrogate)
Intercurrent	treatment policy	6 treated excluded: ICU, death, discharge, stop
Summary	relative risk	proportions, many p-values

OHCA: out-of-hospital cardiac arrest. **Treatment policy**: the outcome counts whatever happens after randomisation (as in intention to treat). **Surrogate**: a biomarker (here a PCR result) used in place of a clinical outcome. **HCQ, AZ**: hydroxychloroquine, azithromycin.

Dankiewicz et al. NEJM 2021; Gautret et al. Int J Antimicrob Agents 2020 (retracted 2024).

Read the interval, not the p-value



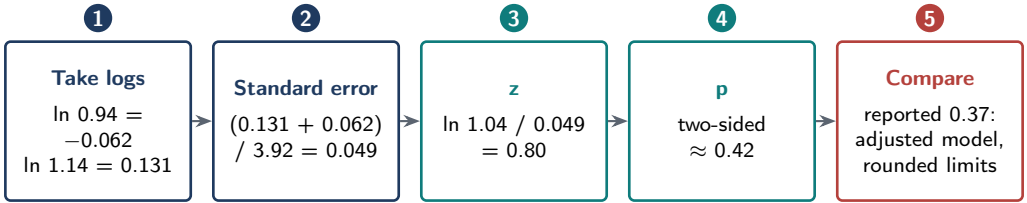
Correct
no difference shown; effects from a 6% reduction to a 14% increase remain compatible

Wrong
"hypothermia does not work"
"the two treatments are equivalent"

RR: relative risk = risk in the treated group / risk in the control group; 1 = no difference. **95% CI**: the range of effects compatible with the data. **Equivalence**: needs a margin fixed in advance and a CI inside it; a non-significant p does not show it.

TTM2: death 465/925 (50%) vs 446/925 (48%).

From a confidence interval back to a p-value



Rule of thumb. A 95% CI that excludes the null implies $p < 0.05$; one that includes it implies $p > 0.05$. A paper that breaks this rule needs a question to the authors.

Logs: ratios (RR, OR, HR) are symmetric on the log scale, so the calculation is done there. **SE**: standard error, the precision of the estimate; a 95% CI is ± 1.96 SE, so its width is 3.92 SE. **z**: estimate / SE; $|z| > 1.96$ means $p < 0.05$.

Altman and Bland, BMJ 2011;343:d2090. Toolkit, layer 4.

Forensics: four checks you can do with a phone

Percentages

recompute every n/N you can find

$87/120 = 72.5\%$, not 75.5%

CI vs p

do they agree?

OR 1.32 (1.01 to 1.73) with p = 0.11: no

GRIM

mean × n must be close to an integer

n = 13, mean 2.41: 31.33: impossible

Flow

every patient from enrolled to analysed

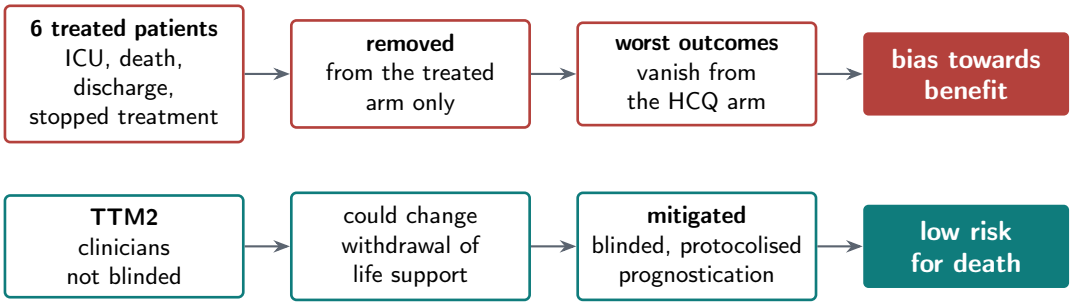
TTM2 adds up; Gautret: no flow diagram

an inconsistency is a question to the authors, not an accusation

GRIM (granularity-related inconsistency of means): with whole-number data, mean × n must give a whole number, allowing for rounding; works for small n. **Flow**: count every patient from enrolled to analysed in each arm; numbers must add up and every loss be explained (CONSORT flow diagram). **OR**: odds ratio.

Brown and Heathers, Soc Psychol Personal Sci 2017 (GRIM); Nuijten et al., Behav Res Methods 2016 (statcheck).

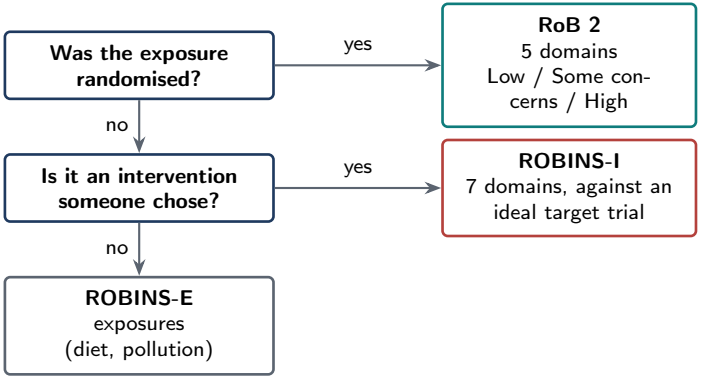
Bias has a direction: ask which way it pushes



Per protocol: analysing only patients who followed the protocol. **Outcome-dependent exclusion**: removing patients because of what happened to them, here the worst outcomes. **Blinded prognostication**: neurological prognosis assessed by someone who did not know the arm.

Calling outcome-dependent exclusions "per protocol" does not remove the bias.

Which risk-of-bias tool?

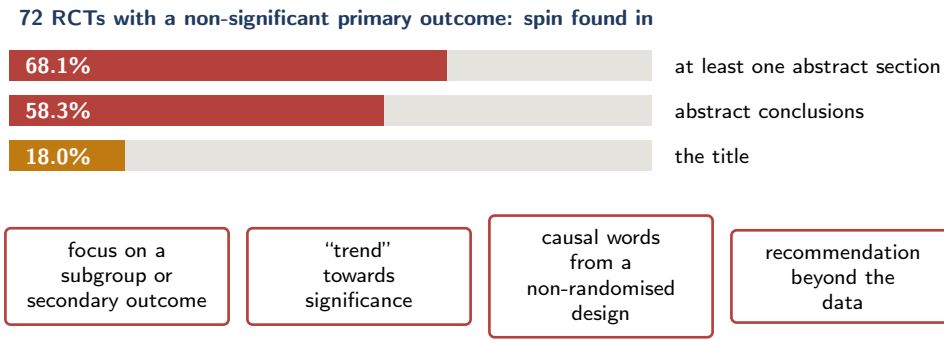


one **Critical** domain makes the whole study
Critical: Gautret (confounding, missing data)

RoB 2: risk-of-bias tool for randomised trials. **ROBINS-I / -E**: for non-randomised studies of interventions / exposures. **Target trial**: the ideal randomised trial the study tries to imitate. **Confounding**: a factor linked to both treatment and outcome.

Sterne et al. BMJ 2019 (RoB 2); BMJ 2016 (ROBINS-I); Higgins et al. Environ Int 2024 (ROBINS-E).

Spin: the abstract says more than the results



Spin: reporting that makes a treatment look better than the results show. **Non-significant primary outcome**: $p \geq 0.05$ on the main pre-specified outcome. **Trend**: wording for a non-significant result presented as almost a benefit.

Boutron et al. JAMA 2010;303:2058.

Causal words must match the design

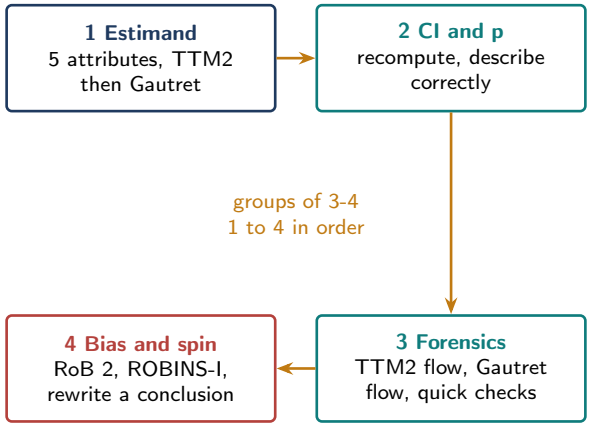
Design	What it can show	Words that fit	Words that overreach
Case report, case series	one observation; a hypothesis	"we observed", "occurred after", "may suggest"	"caused", "induced", "cures", "effective"
Cross-sectional	association at one time point	"was associated with", "more frequent in"	"leads to", "risk factor for"
Cohort, case-control	association over time; confounding remains	"associated with, after adjustment for"	"reduces", "protects", "the effect of"
Target trial emulation	a causal estimate, if its assumptions hold	"estimated effect, assuming no unmeasured confounding"	causal claim with the assumptions left out
RCT, low risk of bias	a causal effect in that population	"reduced", "caused"	"for all patients", beyond the trial

After is not because of. A case report shows a sequence, never a cause. Gautret (non-randomised, 20 vs 16 analysed) writes "its effect is reinforced by azithromycin": a causal claim the design cannot carry.

Post hoc ergo propter hoc: reading a time sequence as a cause. **Target trial emulation**: observational analysis designed to imitate a specific RCT. **Causality scales** (WHO-UMC, Naranjo): grade a drug's role in one patient's adverse event; their strongest word is "probable", never "proven".

Hernán. Am J Public Health 2018;108:616 (the C-word). Naranjo et al. Clin Pharmacol Ther 1981;30:239.

Exercise 2 Four stations, 15 minutes each



Pack p. 6-7

Exercise 2 Four stations — groups of 3-4

4×15

minutes

START NOW

Instructions and answer grid: pack p. 6-7

All groups, in order: $1 \to 2 \to 3 \to 4$.
About 15 minutes per station; I call each change.
Formulas: toolkit, p. 23-25

STOP when time is up → next: debrief (10 min); lunch

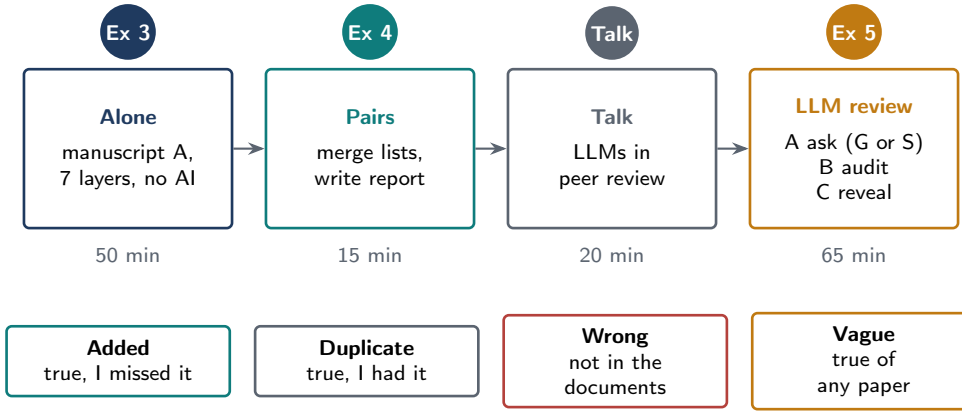
Lunch — 60 minutes

Next: block 3, apply the method to manuscript A

3 Apply the method, then ask an LLM

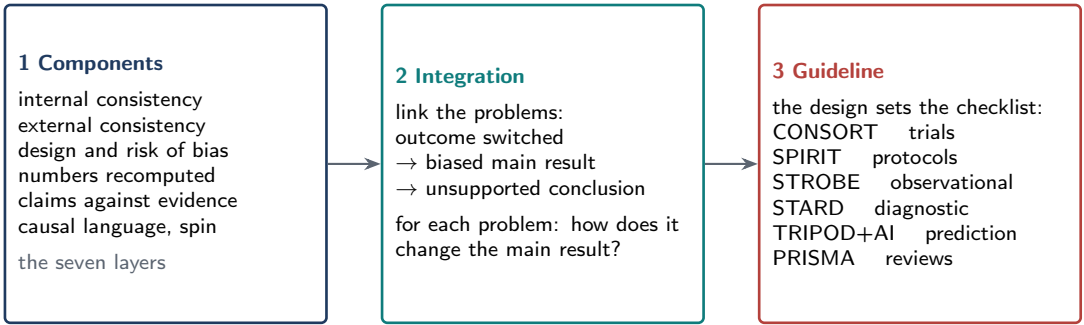
One new paper, reviewed alone with the seven layers, then in pairs.

How the afternoon works



Exercises 3 to 5, pack p. 8-18; llm-exercise/manuscript-A.md. Manuscript A is fictional: never paste a manuscript you review for a journal.

A review is a process: components, integration, guideline



Reporting guidelines: CONSORT randomised trials; SPIRIT trial protocols; STROBE observational studies; STARD diagnostic accuracy; TRIPOD+AI prediction models, including machine learning; PRISMA systematic reviews.

The same mental model is what you give an LLM in prompt S. Links: handouts/toolkit.md, layer 1.

Exercise 3 Review manuscript A

alone, no AI

50

minutes

START NOW

Instructions and answer grid: pack p. 8

Read manuscript A (p. 9-12) with the checklist (p. 19-22), layer by layer.
Fill the table on p. 8: one row per problem, with location, what is wrong (show your arithmetic), why it matters.

STOP when time is up → next: Exercise 4, pairs

The review report: what an author can act on

Summary 3 to 5 neutral sentences: you understood the paper

Major issues at most four, each with three parts → **1 the problem** **2 why it matters** **3 what to do**

Minor issues wording, tables, references

Recommendation and your confidence in the main conclusion

Confidential to the editor concerns you cannot prove

Not: "The statistics are poor." Instead: "Six treated patients are excluded after ICU transfer or death (Section 3.1). Because they are the worst outcomes, this biases the result towards benefit. Please report all 26 treated patients."

Exercise 4 Pair report

pairs

15

minutes

START NOW

Instructions and answer grid: pack p. 13

Merge your two Exercise 3 tables: what did one of you see that the other missed?
Write the report in the template (p. 14-15): summary, major issues, minor issues, recommendation.

STOP when time is up → next: break, then the talk on LLMs

Break — 15 minutes

Next: block 4, LLMs in peer review, then Exercise 5

4 LLMs in peer review: practical concerns

Policies, confidentiality, invented facts, and what the evidence shows. Then Exercise 5: an LLM reviews manuscript A, you audit it, and the reveal.

Where an LLM could enter, and where it must not

Your own paper
before submission:
check reporting

Published papers
screening, extraction
(verified)

allowed, with verification and disclosure

Confidential manuscript
you review for
a journal

do not upload

Your written report
language polishing

only if policy allows

NIH (2023)
reviewers may not use
generative AI on applications

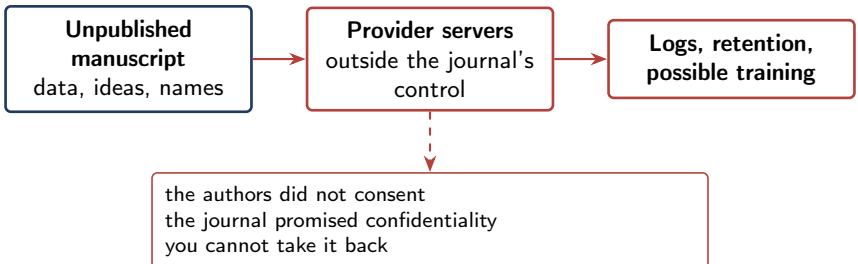
ICMJE
no upload where confidentiality
cannot be assured; disclose use

Most publishers
reviewers must not upload
manuscripts to AI tools

LLM: large language model (ChatGPT, Claude, Gemini, Copilot). **ICMJE**: International Committee of Medical Journal Editors.
Generative AI: AI that produces text or images.

NIH notice NOT-OD-23-149 (2023); ICMJE Recommendations (updated 2024); check your journal's current policy before every review.

Confidentiality: what happens when you paste a manuscript



Today's exercise is safe because manuscript A is fictional and written for training.

Retention: the provider keeps your text for a period. **Training**: the provider may use it to improve future models, so it can resurface.

Four ways an LLM review goes wrong

Invented facts
registry content, protocol details it never saw

Invented refs
plausible authors, journals, years; no such paper

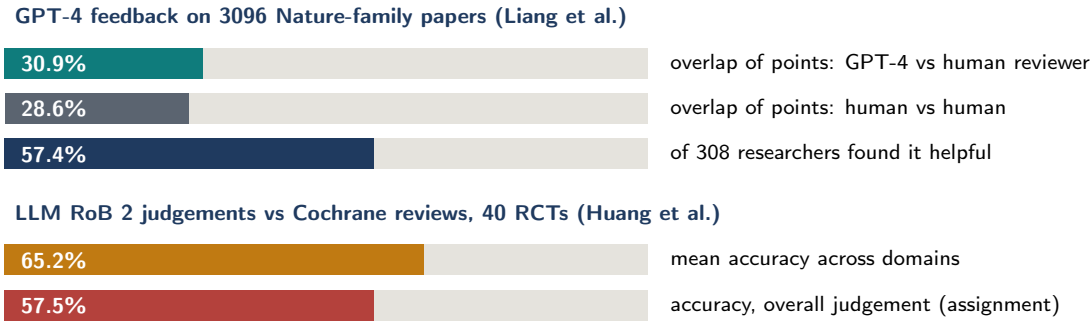
Wrong arithmetic
percentages, p from CI, GRIM, stated confidently

Boilerplate
"consider confounding" for any paper

every LLM claim is a hypothesis you must check in the document

Invented facts ("hallucinations"): fluent, confident statements with no source. **Boilerplate**: generic comments true of any paper.

What the evidence shows so far



Overlap: share of the points raised by one reviewer that the other also raised. **Accuracy**: share of LLM judgements that match the Cochrane review.

Liang et al. NEJM AI 2024; Huang et al. J Med Internet Res 2025;27:e70450 (materials/references/). Useful as a second reader; not a replacement.

When AI conducts research: reproducible is not valid

PNAS 2026: 288 researchers randomised in 103 teams to reproduce published results



An LLM agent on a survival analysis

right analysis plan in 9 of 9 runs, yet systematic code errors repeated across runs; 8 of 17 summaries fully right, 2 clinically wrong.

Three levels

same numbers (reproducible)
$\neq$ right methods (statistically valid)
$\neq$ claim supported (scientifically valid).
Code that runs is only the first level.

AI-assisted: humans lead, AI helps. **AI-led**: AI does the work with little human checking. **LLM agent**: an LLM that writes and runs code by itself.

Brodeur et al. PNAS 2026 (social-science reproduction only); Wu et al. J Med Internet Res 2026 (doi:10.2196/99597).

Two prompts, same paper

Prompt G (generic), odd seats

You are a peer reviewer for a general medical journal. Review the paper below.

Prompt S (structured), even seats

Work through these components in order; each feeds the next:

1 article type and its reporting guideline
2 external consistency with every source given
3 internal consistency: recompute every number
4 design and risk of bias
5 claims and language
6 integration: how each problem changes the conclusion

Location, what is wrong (show arithmetic), why it matters. Rank. Invent nothing.

Both

Paste from "Document 1" to the end. Note the model and its version.

Prompt: the instruction you give the LLM. **Generic**: one sentence, no method. **Structured**: the method of this morning, step by step.

Full text of both prompts: Exercise 5, Part A, pack p. 16.

Exercise 5 LLM review

individual, then plenary

65

minutes

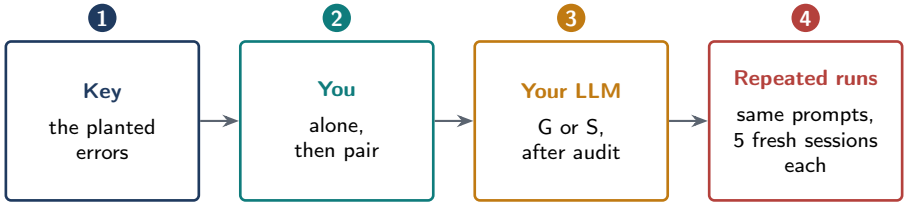
START NOW

Instructions and answer grid: pack p. 16-18

Part A (10 min): odd seat prompt G, even seat prompt S; paste it, then the file llm-exercise/manuscript-A.md from "Document 1".
Part B (20 min): check every LLM point yourself; fill the audit table.
Part C (35 min): reveal with the annotated manuscript A; score table, questions 1-4, your one rule.

STOP when time is up → next: wrap-up and post-quiz

Exercise 5, Part C Reveal and compare



	manuscript only	+ registry and SAP
prompt G	?	? (half the room)
prompt S	?	? (half the room)

Planted errors: errors written into manuscript A on purpose. **Wrong claim**: an LLM point not supported by the documents. **Trap**: a suspicion that only the registry or SAP can confirm.

Errors found, wrong claims, traps flagged. What changes with the prompt? What changes with the sources?

What repeated runs teach, for any paper

Documents beat prompts

a check that needs the registry or SAP cannot be done without it, whatever the prompt

Structure raises the right questions

the structured prompt asked what it could not verify; the generic one mostly did not

Arithmetic is cheap, not free

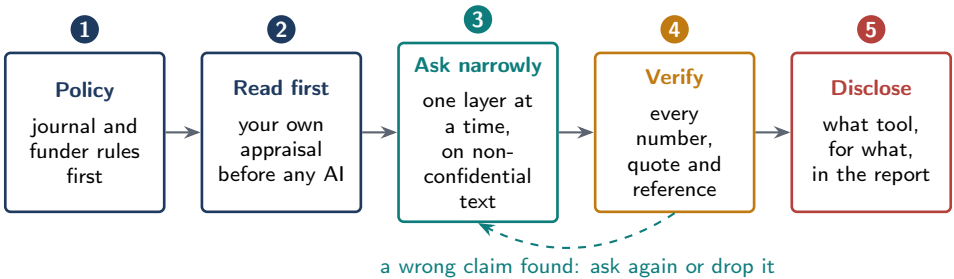
recomputation was near ceiling, yet some runs saw an odd number and did not test it

More text, more to audit

structured reviews were 60 to 80% longer; every point still needs checking

The structure came from the method you learned this morning and from the reporting guideline. Clean planted errors are an easy case: real manuscripts are harder, and a real manuscript under review never goes into an LLM.

A safe routine for AI in your own appraisal work



Wrap-up

Open science
compare the paper with what was promised

Methodology
estimand, interval, numbers, bias direction, spin

Review
problem, why it matters, what to do

LLMs
never confidential text; verify every claim; disclose

Post-quiz
Same 15 questions as the pre-quiz. Which answers changed, and which layer of the method explains the change?

Take-home kit: 7-layer checklist, toolkit, review report template. To go further: the reporting guideline for the designs you read (toolkit, layer 1).

Post-quiz — individual

10

minutes

START NOW

Instructions and answer grid: pack p. 26-27

Same 15 questions as the pre-quiz, one answer each.

STOP when time is up → next: evaluation, and thank you